

第三周作业参考解答

p.44, 习题 11.2, 2(2), (4), 11(3).

$$2. \text{ 解: } (2) \quad \frac{\partial z}{\partial x} = 2xf_1' + ye^{xy} f_2', \quad \frac{\partial z}{\partial y} = -2yf_1' + xe^{xy} f_2';$$

$$(4) \quad \frac{\partial z}{\partial x} = y - \frac{y}{x^2} f(xy) + \frac{y^2}{x} f'(xy), \quad \frac{\partial z}{\partial y} = x + \frac{1}{x} f(xy) + yf'(xy).$$

$$11 (3) . \text{ 证明: } \frac{\partial u}{\partial x} = \frac{-x}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}, \quad \frac{\partial^2 u}{\partial x^2} = \frac{2x^2 - y^2 - z^2}{(x^2 + y^2 + z^2)^{\frac{5}{2}}};$$

$$\frac{\partial u}{\partial y} = \frac{-y}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}, \quad \frac{\partial^2 u}{\partial y^2} = \frac{2y^2 - x^2 - z^2}{(x^2 + y^2 + z^2)^{\frac{5}{2}}};$$

$$\frac{\partial u}{\partial z} = \frac{-z}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}, \quad \frac{\partial^2 u}{\partial z^2} = \frac{2z^2 - x^2 - y^2}{(x^2 + y^2 + z^2)^{\frac{5}{2}}},$$

$$\text{故 } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0. \text{ 证毕}$$

p.50, 习题 11.3, 1, 2(2), 4(3).

1. 解: 因为 $\lim_{(x,y) \rightarrow (0,0)} \sqrt[3]{xy} = 0 = f(0,0)$, 因此 f 在 $(0,0)$ 点连续. 又

$$f'_x(0,0) = \lim_{x \rightarrow 0} \frac{f(x,0) - f(0,0)}{x} = 0, \text{ 同理可求得}$$

$$f'_y(0,0) = \lim_{y \rightarrow 0} \frac{f(0,y) - f(0,0)}{y} = 0, \text{ 故 } f \text{ 在 } (0,0) \text{ 点关于 } x, y \text{ 的偏导数存在.}$$

方向 $\mathbf{v} = (a,b)^T$ 的单位方向向量记为 $\mathbf{e}_v = \left(\frac{a}{\sqrt{a^2+b^2}}, \frac{b}{\sqrt{a^2+b^2}} \right)^T$, 则

$$f'_v(0,0) = \lim_{t \rightarrow 0^+} \frac{f\left(\frac{a}{\sqrt{a^2+b^2}}t, \frac{b}{\sqrt{a^2+b^2}}t\right) - f(0,0)}{t} = \lim_{t \rightarrow 0^+} \frac{\sqrt[3]{t^2 \frac{ab}{a^2+b^2}}}{t}$$

不存在, 故 f 在 $(0,0)$ 点沿方向 $\mathbf{v} = (a,b)^T$ 的方向导数不存在.

2 (2) . 证明: 当 $y \neq 0$ 时, 动点沿曲线 $y = kx^3 (k \neq 0)$ 趋近于坐标原点, $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3}{y}$ 的值与 k 有关, 因此该函数在原点的二重极限不存在, 所以函数 f 在原点处不连续.

下面讨论 f 在原点沿任意方向的方向导数. 不妨设 $\mathbf{l} = (\cos \theta, \sin \theta)$, 当 $\sin \theta = 0$ 时,

$$f'_l(0,0) = \lim_{t \rightarrow 0^+} \frac{f(t \cos \theta, 0) - f(0,0)}{t} = 0;$$

当 $\sin \theta \neq 0$ 时, $f'_1(0,0) = \lim_{t \rightarrow 0^+} \frac{f(t \cos \theta, t \sin \theta) - f(0,0)}{t} = \lim_{t \rightarrow 0^+} \frac{t \cos^3 \theta}{\sin \theta} = 0$, 因此 f 在原点沿任何方向的方向导数都存在, 且都等于零.

4. (3) 由 $df(x_0, y_0) = 3dx - 2dy$ 可知 $f'_x(x_0, y_0) = 3$, $f'_y(x_0, y_0) = -2$. 又该函数在 (x_0, y_0) 点沿 I_1 方向增长最快, 沿 I_2 方向减小最快, 故 I_1 是梯度方向且 $I_1 = \frac{1}{\sqrt{13}}(3, -2)$, I_2 为梯度反方向且 $I_2 = \frac{1}{\sqrt{13}}(-3, 2)$.

p.64, 习题 11.5, 2(1), 3(1), 4(1), 5.

2 (1). 解: 令 $F(x, y, z) = x + y - z - \cos(xyz)$, 因为

$$\frac{\partial F}{\partial z}(0, 0, -1) = -1 + xy \sin(xyz) \Big|_{(0,0,-1)} = -1 \neq 0,$$

因此由方程 $x + y - z - \cos(xyz) = 0$ 确定了一个由 $(0,0)$ 的邻域到 -1 邻域的隐函数

$$z = z(x, y), \text{ 且 } \frac{\partial z}{\partial x} = \frac{1 + yz \sin(xyz)}{1 - xy \sin(xyz)}, \quad \frac{\partial z}{\partial y} = \frac{1 + xz \sin(xyz)}{1 - xy \sin(xyz)}.$$

3 (1). 解: 令 $F(x, y, z) = e^{-(x+y+z)} - (x + y + z)$, 且 $f(t) = e^{-t} - t$, ($t > 0$).

由于 $f'(t) = -e^{-t} - 1 < 0$, ($\forall t > 0$), 因此 $f(t) = e^{-t} - t = 0$ 有唯一的解 t_0 , 故平面

$x + y + z = t_0$ 上的点都是方程 $e^{-(x+y+z)} = (x + y + z)$ 的解, 且在该平面上的每一点处, 都有

$F'_z(x, y, z) = -e^{-(x+y+z)} - 1 \neq 0$, 所以方程 $e^{-(x+y+z)} = (x + y + z)$ 在平面 $x + y + z = t_0$ 上的

任一点附近都可以确定隐函数 $z = z(x, y)$, 且 $\frac{\partial z}{\partial x} = -\frac{F'_x}{F'_z} = -1$; $\frac{\partial z}{\partial y} = -\frac{F'_y}{F'_z} = -1$.

4 (1). 解: 方程 $f(ax - cz, ay - bz) = 0$ 两边对 x 求偏导, 得 $(a - c \frac{\partial z}{\partial x}) f'_1 - b \frac{\partial z}{\partial x} f'_2 = 0$, 解得

$$\frac{\partial z}{\partial x} = \frac{af'_1}{cf'_1 + bf'_2}, \text{ 同理求得 } \frac{\partial z}{\partial y} = \frac{af'_2}{cf'_1 + bf'_2}, \text{ 故 } c \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} = \frac{caf'_1 + baf'_2}{cf'_1 + bf'_2} = a.$$

5. 解: 记 $F_1 = x + y + z + z^2$, $F_2 = x + y^2 + z + z^2$. 则

$$\frac{\partial(F_1, F_2)}{\partial(y, z)} = \begin{pmatrix} 1 & 1+2z \\ 2y & 1+3z^2 \end{pmatrix}, \quad \frac{\partial(F_1, F_2)}{\partial x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

因为 $F_1(-1, 1, 0) = F_2(-1, 1, 0) = 0$, 且 F_1, F_2 均在点 $(-1, 1, 0)$ 的邻域内连续, 也有

$$\frac{\partial(F_1, F_2)}{\partial(y, z)} \bigg|_{(-1, 1, 0)} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \text{可逆},$$

因此能在 $P(-1, 1, 0)$ 附近确定 $\begin{pmatrix} y \\ z \end{pmatrix} = \mathbf{f}(x)$. 由于

$$\frac{\partial(y, z)}{\partial x} = - \left(\frac{\partial(F_1, F_2)}{\partial(y, z)} \right)^{-1} \frac{\partial(F_1, F_2)}{\partial x} = - \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix},$$

$$y'(-1) = 0; \quad z'(-1) = -1.$$

p.64, 习题 11.5, 6(1), 7, 8, 10, 14.

6 (1). 解: 等式两边对 x 求导, 有 $f_1' \left(1 + \frac{\partial z}{\partial x} \right) + f_2' \left(2x + 2z \cdot \frac{\partial z}{\partial x} \right) = 0$, 所以

$$\frac{\partial z}{\partial x} = - \frac{f_1' + 2x \cdot f_2'}{f_1' + 2zf_2'}; \text{同理, 等式两边对 } y \text{ 求导 } f_1' \left(1 + \frac{\partial z}{\partial y} \right) + f_2' \left(2y + 2z \cdot \frac{\partial z}{\partial y} \right) = 0, \text{ 得}$$

$$\frac{\partial z}{\partial y} = - \frac{f_1' + 2y \cdot f_2'}{f_1' + 2zf_2'}, \text{ 故 } dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = - \frac{f_1' + 2x \cdot f_2'}{f_1' + 2zf_2'} dx - \frac{f_1' + 2y \cdot f_2'}{f_1' + 2zf_2'} dy.$$

7. 解: 令 $F(x, y, z) = x^3 + y^3 + z^3 - \varphi(z)$, 则当 $\frac{\partial F}{\partial z} = 3z^2 - \varphi'(z) \neq 0$ 时, 由方程

$x^3 + y^3 + z^3 = \varphi(z)$ 能确定隐函数 $z = z(x, y)$. 在方程 $x^3 + y^3 + [z(x, y)]^3 = \varphi(z(x, y))$ 两边分别对变量 x, y 求偏导, 得

$$3x^2 + 3z^2 \frac{\partial z}{\partial x} = \varphi'(z) \frac{\partial z}{\partial x},$$

$$3y^2 + 3z^2 \frac{\partial z}{\partial y} = \varphi'(z) \frac{\partial z}{\partial y}.$$

从而 $\frac{\partial z}{\partial x} = \frac{3x^2}{\varphi'(z) - 3z^2}$, $\frac{\partial z}{\partial y} = \frac{3y^2}{\varphi'(z) - 3z^2}$. 因为 φ 二阶连续可微, 因此 $z = z(x, y)$ 二阶连

续可微, 对 $\frac{\partial z}{\partial x} = \frac{3x^2}{\varphi'(z) - 3z^2}$ 关于 y 求偏导, 有 $\frac{\partial^2 z}{\partial x \partial y} = \frac{9x^2 y^2 (6z - \varphi''(z))}{(\varphi'(z) - 3z^2)^3}$.

8. 证明: 方程 $F(x + \frac{u}{y}, y + \frac{u}{x}) = 0$ 两边分别对 x, y 求偏导, 有

$$F_1' \left(1 + \frac{1}{y} \frac{\partial u}{\partial x} \right) + F_2' \left(\frac{1}{x} \frac{\partial u}{\partial x} - \frac{u}{x^2} \right) = 0 \text{ 且 } F_1' \left(\frac{1}{y} \frac{\partial u}{\partial y} - \frac{u}{y^2} \right) + F_2' \left(1 + \frac{1}{x} \frac{\partial u}{\partial y} \right) = 0,$$

$$\text{所以 } \frac{\partial u}{\partial x} = \frac{\frac{u}{x^2} F_2' - F_1'}{\frac{F_1'}{y} + \frac{F_2'}{x}}, \quad \frac{\partial u}{\partial y} = \frac{\frac{u}{y^2} F_1' - F_2'}{\frac{F_1'}{y} + \frac{F_2'}{x}}, \quad \text{故 } x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = u - xy. \quad \text{证毕}$$

10. 解: 将 z 看成以 w 为中间变量, 以 x, y 为自变量的复合函数, 其中 $w = w(u, v)$,

$u = x, v = x + y$, 则

$$\frac{\partial z}{\partial x} = \frac{\partial w}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial w}{\partial v} \frac{\partial v}{\partial x} - 1 = \frac{\partial w}{\partial u} + \frac{\partial w}{\partial v} - 1,$$

$$\frac{\partial z}{\partial y} = \frac{\partial w}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial w}{\partial v} \frac{\partial v}{\partial y} - 1 = \frac{\partial w}{\partial v} - 1,$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 w}{\partial u^2} \frac{\partial u}{\partial x} + \frac{\partial^2 w}{\partial u \partial v} \frac{\partial v}{\partial x} + \frac{\partial^2 w}{\partial u \partial v} \frac{\partial u}{\partial x} + \frac{\partial^2 w}{\partial v^2} \frac{\partial v}{\partial x} = \frac{\partial^2 w}{\partial u^2} + 2 \frac{\partial^2 w}{\partial u \partial v} + \frac{\partial^2 w}{\partial v^2},$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 w}{\partial u \partial v} \frac{\partial u}{\partial x} + \frac{\partial^2 w}{\partial v^2} \frac{\partial v}{\partial x} = \frac{\partial^2 w}{\partial u \partial v} + \frac{\partial^2 w}{\partial v^2},$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial^2 w}{\partial u \partial v} \frac{\partial u}{\partial y} + \frac{\partial^2 w}{\partial v^2} \frac{\partial v}{\partial y} = \frac{\partial^2 w}{\partial v^2},$$

将其代入方程 $\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} + \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} = 0$ 中, 得方程 $\frac{\partial^2 w}{\partial u^2} + \frac{\partial w}{\partial u} = 0$.

14. 解: 由于 $\left. \frac{\partial(x, y)}{\partial(u, v)} \right|_{(2,1)} = \begin{pmatrix} 1 & 1 \\ 2uv^2 & 2u^2v \end{pmatrix} \bigg|_{(2,1)} = \begin{pmatrix} 1 & 1 \\ 4 & 8 \end{pmatrix}$ 可逆, 故

$$\left. \frac{\partial(u, v)}{\partial(x, y)} \right|_{(3,4)} = \left(\left. \frac{\partial(x, y)}{\partial(u, v)} \right|_{(2,1)} \right)^{-1} = \begin{pmatrix} 2 & -\frac{1}{4} \\ -1 & \frac{1}{4} \end{pmatrix}, \quad \text{从而 } \frac{\partial u}{\partial x}(3, 4) = 2.$$